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Project Title:	Effects of Estrogen on Cataract Induction After Exposure to High LET Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	46202-5121	**Congressional District:**	7
Comments:			
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 5/31/2011 per C. Guidry/JSC (10/2010) NOTE: PI to receive NCE to 8/31/2010 per J. Dardano/JSC (7/09)		
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):	Mendonca, Marc (Indiana U) Valluri, Shelia (Indiana U) Bigsby, Robert (Indiana U)		
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Task Description:	Several studies have demonstrated that there is an increased cataratogenic potential in animals and humans exposed to high linear energy transfer (LET) radiation compared to low LET radiation. Since the relative biological effectiveness (RBE) of heavy energetic charged (HZE) particles for lens opacification is high, these studies imply a high cataract risk for astronauts involved in long-duration space missions. During a manned mission to Mars, ocular doses in excess of ~0.1 Gy of HZE particles could be accumulated. Though not directly life-threatening, radiation-induced cataractogenesis represents a potentially serious sequelae which could necessitate surgical intervention in the years after, or perhaps even during the mission in some individuals (see below). While the cellular and molecular mechanism(s) of radiation-induced cataractogenesis have not been clearly elucidated, damage to DNA at the time of exposure and subsequent proliferation of the radiosensitive cells in the germinative zone of the lens epithelium likely play a role in the process. Using a rat model, we have recently accumulated data which indicate that estrogen increases or reduces the latent period and may modulate the incidence and severity of cataracts induced by low LET radiation, depending upon when the estrogen is administered. High estrogen levels are induced in non-pregnant women using oral contraceptives, or in post-menopausal women on estrogen replacement therapy. Therefore, sub-groups of female astronauts may be at an increased risk for developing cataracts that are more severe or occur with a more rapid onset. The interaction between estrogen and high LET radiation has never been studied before in the context of cataractogenesis. Therefore, a mechanistic approach characterizing the modulating effect of estrogen on high LET radiation-induced cataractogenesis is proposed. The underlying hypothesis of this proposal is that estrogen alters the latent period for high LET radiation-induced cataractogenesis, and, depending upon when the it is administered, modulates severity and incidence of cataracts. Estrogens regulate several proteins involved in cell cycle control and apoptosis, and it's metabolism results in the production of free radicals which may be genotoxic and mutagenic to mammalian cells. Thus, a novel hypothesis to be tested in the proposed studies is that estrogen alters cell cycle regulation, DNA double strand break (DSB) induction or repair, and proliferation in lens cells exposed to HZE particles (1 GeV/amu iron ion beams). The first Specific Aim shall be to define the dose-time interactions of high LET radiation (vs. low LET radiation) and estradiol to better understand the mechanism of estrogen action. The RBE for cataractogenesis in estrogen-treated animals exposed to1 GeV iron beams shall be determined using 60-Co gamma rays as a standard. The second specific aim shall be to determine the mechanism of estrogen-modulation of radiation-induced cataracts. Animal and cell culture studies will be performed to determine the effects of estrogen on gene expression, cell proliferation, apoptosis, and the induction and repair of radiation-induced DSBs. Slit-lamp biomicroscopy will be used to measure lens opacification in exposed animals. Successful completion of these studies will lead to a greater understanding of the modulation of space radiation effects by hormones and novel intervention strategies for long-duration space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results obtained from this project may provide greater insight into the potential effects of hormones on normal tissues after radiotherapy treatments with high LET particles.
Task Progress:	This report summarizes progress during the fourth year of support for NASA Grant NNJ05HE64G. While work mainly was focused on performing experiments to complete the first Specific Aim of the project [to determine whether estrogen (17-beta-estradiol, E2) alters cataractogenesis induced by high LET radiation], we also performed experiments relevant to the second Specific Aim (mechanisms of modulation of cataractogenesis by E2). During the fourth year of funding, we completed our fifth "run" at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL); beam time had been requested in the Spring 2009 run. In all of our runs to date, we have used our specially designed animal holder and collimator set-up to irradiate the right eyes of up to 3 rats simultaneously, while sparing the contralateral eye and most other normal tissues. This resulted in other structures receiving a dose of <3.5% of the dose to the target eye. We have now completed measurements of opacification in over three hundred rats that have been irradiated in this manner, and we have yet to observe a radiation-induced cataract in the contralateral eye. This validates the use of our animal holder and collimator set-up for single eye irradiations, and also suggests that the devices can be used to target small volumes of tissue in animal models. We completed our accrual of data from an experiment that was initiated during Year 2, in which rats were irradiated during our second run at NSRL in 2006. Male and female rats were exposed to 1 Gy of a 600 MeV/amu 56-Fe ion beam in order for us to assess gender-based differences in cataractogenesis after exposure to high LET radiation. Eyes of placebo-implanted or E2-implanted 56-day old male rats and ovary-intact placebo-implanted 56-day old female rats were irradiated and shipped back to Indiana University. Slit-lamp biomicroscopy was used to measure lens opacification in exposed animals. Opacification of the anterior and posterior subcapsular (ASC and PSC, respectively) lenses of irradiated and control (contralateral) eyes was measured every 2-4 weeks. In addition to permitting an assessment of gender differences for cataractogenesis, after high LET irradiation, this experiment also allowed us to determine whether estrogen would modulate radiation cataractogenesis. Since the incidence of cataracts (either ASC or PSC) was significantly increased in intact female rats and male rats with E2, compared to male rats with empty capsules implanted, we concluded that either exogenous or endogenous estrogen enhanced cataractogenesis. This result was contradictory to what we observed in similar experiments conducted with low LET radiation, the results of which were presented in a previous task report. In the low LET experiments, exogenous estrogen did not appear to account for gender differences, although endogenous estrogen clearly enhanced cataractogenesis. Beginning during the Fall 2006 run and continuing through the Fall 2007 run, a radiation dose response experiment was performed at NSRL to determine whether modulation of cataractogenesis by E2 after exposure to high LET radiation is dependent upon the radiation dose. Ovariectomized or intact female rats, or ovariectomized rats that received estrogen-implants were irradiated with 0.05 to 2 Gy of 600 MeV/amu 56Fe ions. All doses have now been delivered, and although data accrual is almost complete for the 1 and 2 Gy groups (since latent period for cataract formation is inversely related to dose), data accrual is continuing for animals irradiated with lower doses. At this time though, it does appear that estrogen enhances cataractogenesis to a greater extent after exposure to lower doses of iron ions. Results with 1 and 2 Gy doses have yielded interesting results. The rate of PSC progression after irradiation with 1 Gy of iron ions was significantly greater in intact females and ovariectomized females with E2, compared to ovariectomized females with empty capsules implanted (ASC & PSC progression was similar). Also, the incidence of cataracts (either ASC or PSC) was significantly higher in intact females and ovariectomized females with E2, compared to ovariectomized females with empty capsules implanted. Thus, it appears that endogenous or exogenous estrogen may enhance both the rate and incidence of cataractogenesis after exposure to high LET radiation. After irradiation with 2 Gy of iron, the rate of PSC progression was significantly greater in ovariectomized females implanted with E2 compared to

	ovariectomized or intact animals (ASC & PSC progression was again similar). Ovariectomized females with implanted E2 also had a significantly higher incidence of cataract (either ASC or PSC) than the other groups. We conclude that enhancement of radiation cataractogenesis by estrogen appears be dependent upon LET as well as dose. Additional experiments were performed during the Fall 2007 and Spring 2008 runs to determine the effects of age on radiation cataractogenesis induced by high LET radiation, as well as the effects of timing of the estrogen response. In this latter experiment, the goal was to determine whether estrogen, if implanted after irradiation, can protect against high LET radiation-induced cataractogenesis, as we have observed after exposure to low LET radiation. We have found that only minimal protection is afforded if estrogen is removed after irradiation, but no protection is afforded if it is added after irradiation. In regard to the age response, we have found that E2 enhances cataractogenesis is old or young animals irradiated with iron ions, but older animals show a greatly reduced latent period and incidence compared to younger animals. When data accrual for all of these experiments is complete, the issue of age and gender differences in radiation cataractogenesis, and the issue of whether hormones may be determinants, will be better understood. These data may have important implications in regard to age and gender consideration for extended manned space flight.
Bibliography Type:	Description: (Last Updated: 07/17/2025)
Abstracts for Journals and Proceedings	Valluri S, Garrett J, Mendonca MS, Bigsby RM, Caperell-Grant A, Rusek A, Dynlacht JR. "Age as a determinant in the modulation of radiation-induced cataractogenesis by estrogen." Presented at the Association for Research in Vision and Ophthalmology, Fort Lauderdale, FL, May 2009. Association for Research in Vision and Ophthalmology, Fort Lauderdale, FL, May 2009. , May-2009
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Articles in Peer-reviewed Journals	Henderson MA, Valluri S, Garrett J, Lopez JT, Caperell-Grant A, Mendonca MS, Rusek A, Bigsby RM, Dynlacht JR. "Effects of estrogen and gender on cataractogenesis induced by high linear energy transfer radiation." Radiation Research, submitted July 2009. , Jul-2009